Co1-At3

Renusree
192511035

Short Answers

1) Tasks A and B share CAN and SPI Flash with opposite acquistion order, reaching a petri Net deadlock. Identify the deadlock marking. Prove the violated Property, and evaluate recourse ordering as Prevention.

Deadlock Marking:
- Task A holds CAN and waits for SPI Flash.
- Task B holds SPI Flash and waits for CAN.
- Both tasks wait forever, causing a deadlock.

Violated property:
- The Liveness Property is violated because no transition can execute once the deadlock occurs.

Resource ordering Prevention:
- Make both tasks acquire resources in the same order (CAN → SPI Flash)
- This removes the circular wait condition and Prevents deadlock while allowing safe resource sharing.

2) Three CAN nodes transmit IDs 0x1A4, 0x1A0, and 0x1B0 simultaneously at 500kbps. Perform arbitration, identify the winner, and calculate arbitration latency in microseconds
Given
→ CAN IDs: 0x1A4, 0x1A0, 0x1B0
→ Bit rate = 500kbps

1. Arbitration: In CAN, the lowest ID has the highest Priority.
   - 0x1A0 = 416
   - 0x1A4 = 420
   - 0x1B0 = 432

2. Arbitration Latency: At 500 kbps,

- Time for 1 bit = 1 / 500,000 = 2 μs

The IDs differ at the 8th arbitration bit, so arbitration completes in about:

Latency = 8 × 2 μs = 16 μs

3) CAN FD (5Mbps) passes bench test but shows 3% CRC errors in vehicle due to 15cm stubs and ± 20% termination tolerance. Analyse the signal integrity failure and Propose two ISO 11898-2 countmeasures.

→ At 5Mbps, 15cm stubs cause signal reflections

→ ± 20% termination tolerance creates impedance mismatch

→ These effects distort the CAN signal, resulting in 3% CRC errors.

ISO 11898-2 Countermeasures:

1. Reduce stub length to minimize reflections.
2. use Proper 120 Ω termination resistors with tight tolerance at both ends of the CAN bus to ensure correct impedance matching.

4) A Predictive Health Assessment node runs ML inference (1000 MAC's at 50/sec) on Cortex-M4 at 100 MHz (2cycles/MAC). calculate CPU utilisation and available cycles for CAN and Sensor tasks.

Given:

- ML inference = 1000 MACs
- Rate = 50 inferences / Sec
- Cortex-M4 = 100 MHz
- 2 cycles / MAC

→ Cycles per inference = 1000 × 2 = 2000 cycles

→ Cycles per second = 2000 × 50 = 100,000 cycles / sec

→ Total CPU cycles / sec = 100 MHZ = 100,000,000 cycles / sec

CPU utilisation:

$$\frac{100,000}{100,000,000} \times 100 = 0.1\%$$

Available cycles for CAN and sensor tasks:

- 100,000,000 − 100,000 = 99,900,000 cycles / sec

5) An automotive embedded system must satisfy ISO 26262 ASIL-D, CAN FD at 5 Mbps for ADAS, and 10 year component availability simultaneously. Analyse the competing constraints and justify architecture trade-offs.

Competing constraints:

- ISO 26262 ASIL-D: Requires high functional safety, redundancy, and fault detection.
- CAN FD (5 Mbps): Needs high-speed, reliable communication for ADAS with good signal integrity.
- 10 year availability: Requires automotive-grade components with long-term manufacturer support.

Architecture Trade-offs:

→ Use an ASIL-D certified MCU with safety features

→ Use CAN FD transceivers with proper 120 Ω termination and short stub lengths for reliable 5 Mbps communication

→ Select automotive-qualified components (AEC-Q100) from vendors offering 10-year product availability.